

Final Executive Summary: TS-QFormer for Time-Series Question Answering

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1 Introduction

Natural-language question answering (QA) over time series aims to enable users to interact with temporal data using free-form queries such as “When does the first arrhythmia occur?” or “Is the heart rate increasing after 30 seconds?”. This capability is increasingly important in domains such as healthcare and physiological monitoring, where temporal patterns are complex and where practitioners benefit from intuitive, interpretable interfaces. Despite substantial progress in large language models (LLMs), these models cannot directly process long, continuous, high-frequency signals that lack a natural tokenization structure. Conversely, traditional time-series models are specialized for forecasting or classification, not open-ended reasoning.

Bridging this gap requires a mechanism that can distill the relevant temporal structure into a compact representation that aligns with the semantics of a natural-language question. Recent multimodal frameworks highlight the difficulty of this alignment: static embeddings often fail to capture localized physiological events [9]. A very recent work, OpenTSLM [5], achieves SOTA performance on time-series QA reasoning, but it is restricted to domains where large scale training data is available.

In this work, we aim to address the situation where large domain data may not exist. This is especially common for scientific domains such as biology, clinical trials and nuclear fusion research, where data is scarce. Thus, to be able to utilize models in these settings, we need efficient architectures which are able to leverage pre-trained models and are adaptable to these low data-domain specific settings. Motivated by these limitations, we propose **TS-QFormer**, a parameter-efficient adaptor that performs question-conditioned querying over temporal embeddings, producing evidence that a frozen LLM can reason over.

At a high level, TS-QFormer adapts the Q-Former architecture from vision-language modeling to the time-series domain. It uses learnable query tokens guided by the question to selectively extract temporal features from a frozen encoder. The resulting representation is then passed to a frozen language model to generate answer predictions. This work builds upon BLIP2 architecture [6] and [9]. We introduce linear projection layers to merge channel embeddings and also condition the Q-former on the question being asked, allowing it to generate context specific query vectors. We then test performance over in-domain dataset (Human Activity Recognition) and out-of-domain dataset (ECG-QA).

2 Task, Dataset, and Evaluation Metrics

Task Definition. We study the problem of natural-language question answering over multivariate time series. Let $X = \{x_1, \dots, x_T\}$ denote a d -dimensional temporal signal, where each observation $x_t \in R^d$ is a physiological measurement at time t . Given a natural-language query Q , the objective is to produce a short-form answer A , optionally accompanied by a chain-of-thought rationale. Formally, we learn a mapping

$$f : (X, Q) \rightarrow A,$$

where f must extract temporally relevant evidence from X , align this evidence with the semantics of Q , and generate an answer that faithfully reflects the underlying temporal patterns. This setting requires reasoning across multiple temporal scales, detecting significant events, and grounding continuous signals in discrete linguistic concepts.

Datasets. Our primary benchmark is the HAR-QA-CoT dataset introduced in the OpenTSLM framework, which pairs multivariate human-activity-recognition (HAR) time series with templated questions, short natural-language answers, and synthetic chain-of-thought rationales. Each example consists of (i) a short window of

wearable-sensor time series, (ii) a question about the underlying activity pattern (e.g., activity type, transitions, duration, or relative intensity), (iii) a categorical or short free-form textual answer, and (iv) an associated CoT explanation. This setup makes HAR-QA-CoT a natural testbed for evaluating whether TS-QFormer can align time-series representations with language and answer fine-grained, activity-centric questions.

We use HAR-QA-CoT for both training and in-domain evaluation, following the official train/validation/test splits provided by OpenTSLM to enable controlled comparison against their baseline models. To assess out-of-domain generalization, we additionally evaluate the same models on ECG-QA-CoT [5] *without* further fine-tuning on ECG data. ECG-QA-CoT contains QA-CoT annotations over short 12-lead ECG segments and differs substantially from HAR in modality, sampling characteristics, and question semantics. This cross-dataset setting allows us to probe whether temporal abstractions learned from wearable-activity signals transfer to a clinically oriented physiological domain. Together, these datasets are well matched to our goal of evaluating both in-domain temporal reasoning and robustness under significant domain shift.

Evaluation Metrics. For both HAR-QA-CoT (in-domain) and ECG-QA-CoT (out-of-domain), we evaluate models along two complementary dimensions: answer correctness and textual similarity. First, we report answer accuracy, defined as the proportion of examples where the model’s predicted answer exactly matches the reference answer after standard normalization (e.g., lowercasing and stripping punctuation). This metric directly reflects task performance in the underlying QA setting. Second, we compute ROUGE-1 F1 scores between the predicted and reference answers (and short rationales when available). The ROUGE metric captures unigram similarity, providing a more graded view of rationale alignment than strict accuracy. In our analysis, we use answer accuracy as the primary metric and ROUGE scores to characterize qualitative differences in answer fluency and partial correctness across models and domains.

3 Related Work

Research connecting time-series modeling and natural-language reasoning has accelerated, producing several lines of work relevant to TS-QFormer. In multimodal alignment, BLIP-2 [6] introduced the Q-Former, showing that a small set of learnable queries can efficiently bridge frozen encoders and LLMs. Its success in vision–language systems motivates adapting query-based mechanisms to continuous temporal signals, where selective and parameter-efficient evidence extraction remains a core challenge.

In time-series–language integration, recent methods explore joint temporal–textual representation learning. ITFormer [11] proposes a shared embedding space between time-series encoders and LLMs, while ChatTime [10] develops a multimodal foundation model unifying numerical and textual reasoning. Broad benchmarks such as MTBench [2] and TimeSeriesExam [1] evaluate temporal understanding but do not target question-conditioned evidence extraction or clinical reasoning.

Time-series QA has been shaped by Time-MQA [3], which unifies numerical, classical, and open-ended tasks under a QA interface. OpenTSLM [5] extends this with domain-specific CoT datasets (HAR-CoT, Sleep-CoT, ECG-QA-CoT) and highlights both the promise and limits of soft prompts and cross-attention modules for temporal–text alignment. MetaQueries [8] emphasizes hierarchical, multi-resolution querying for structured temporal extraction.

Work on temporal reasoning further motivates our approach. Kong et al. [4] argue for architectures capable of grounded temporal reasoning in multimodal LLMs, while Liu et al. [7] show that visualization-based representations can support temporal pattern interpretation. Collectively, these studies highlight the difficulty of aligning continuous dynamics with linguistic semantics and motivate architectures that are parameter-efficient, question-aware, and multi-resolution.

Even though these works present significant advancements, our problem setting aims to analyze cases where domain-specific paired time-series and text data exist only in limited amounts. Thus, we study how pretrained models (time-series foundational models and LLMs) can be adapted for this multimodal task using lightweight adaptors. The method most closely related to ours is JoLT [9], which also utilizes a Q-Former and summarizes

clinical signals before LLM decoding; however, JoLT does not take full advantage of multi-channel time-series structure and treats time-series analysis primarily as a text summarization task.

Our method, TS-QFormer, builds on these insights by adapting Q-Former mechanisms to time series, enabling selective, question-conditioned extraction from frozen temporal encoders.

4 Methods

TS-QFormer is a parameter-efficient adaptor that inserts a question-conditioned querying module between a frozen time-series encoder and a frozen large language model. This design enables natural-language reasoning over continuous signals while keeping all backbone parameters fixed. The architecture contains three components: (i) a pretrained MOMENT encoder that provides temporal embeddings, (ii) a Q-Former-style adaptor that performs question-aware evidence extraction, and (iii) a frozen OPT-2.7B decoder that generates rationales and answers.

Encoder and channel projection. We use the MOMENT-large encoder, configured as in `moment_har_cot.yaml`, which tokenizes each multivariate time series into patched temporal tokens. The encoder outputs per-channel embeddings

$$H \in R^{T \times d_h}.$$

As specified in `qformer_har_cot.yaml`, we employ a learned linear projection to aggregate channel information:

$$H'_t = W_{\text{chan}} H_t + b_{\text{chan}}, \quad H' \in R^{T \times d_z},$$

where $d_z = 1024$. The MOMENT backbone remains frozen; only the projection parameters are trained.

Question-conditioned querying. The adaptor maintains $K = 64$ learnable query tokens. To incorporate question semantics, the question Q is first embedded by the text encoder associated with the Q-Former module, and the queries interact with these embeddings through self-/cross-attention. Each conditioned query $\tilde{q}_i$ then performs guided cross-attention over temporal embeddings H' :

$$\alpha_{i,t} = \text{softmax}\left(\left(W_q \tilde{q}_i\right)^\top (W_k H'_t)\right), \quad \hat{z}_i = \sum_{t=1}^T \alpha_{i,t} W_v H'_t,$$

producing

$$Z = [\hat{z}_1, \dots, \hat{z}_{64}] \in R^{64 \times 1024}.$$

This mechanism implements selective, question-conditioned temporal evidence extraction. All projections W_q, W_k, W_v are trainable; encoder and decoder weights remain fixed throughout.

LLM integration. The summaries Z are projected by a learned linear map into the OPT-2.7B embedding space and concatenated with the token embeddings of the question to form the decoder input. No LoRA or other parameter-tuning mechanism is applied to the LLM; the decoder remains entirely frozen, ensuring parameter efficiency and isolating learning to the adaptor.

Training objectives. Following the Generative Learning (GL) configuration (`har_cot_gl.yaml`), training proceeds in two stages. **(1) Alignment.** A BLIP-2-style alignment phase grounds temporal summaries in question semantics using contrastive and matching objectives. **(2) Supervised generation.** The adaptor is then optimized to generate chain-of-thought rationales and final answers via standard cross-entropy over output tokens. Since both the encoder and LLM remain frozen, all learning occurs within the adaptor and channel-projection layers. Training hyperparameters (batch size, learning rate schedule, warmup steps, and number of epochs) follow the exact values specified in the provided YAML configurations without modification.

Baselines. We compare TS-QFormer against three parameter-efficient baselines implemented under identical preprocessing and compute conditions: OpenTSLM-SoftPrompt, which injects learned soft prompts into a frozen LLM; OpenTSLM-Flamingo, which adds Flamingo-style cross-attention between temporal features and the LLM; and JoLT, which uses fixed (question-agnostic) Q-Former query tokens. These baselines isolate the contributions of question conditioning, channel-aware projection, and selective temporal querying in TS-QFormer.

5 Experiments and Results

Experimental Setup. We train TS-QFormer on the HAR-QA-CoT dataset, updating only the adaptor and projection layers while keeping both the pretrained time-series encoder and the LLM decoder frozen. This follows the parameter-efficient alignment philosophy of OpenTSLM but replaces their cross-attention block with a question-conditioned Q-Former module. All models are trained on the official HAR-QA-CoT training split and evaluated on the held-out test split. An example of the prompt and response is shown in the Appendix A. The response is a reasoning text paragraph followed by an 'Answer'. We report two main metrics: (i) answer accuracy (exact match after simple normalization) and (ii) ROUGE-F1 between the predicted and reference answer strings. During development, we also monitored ROUGE-1/ROUGE-2 to sanity-check that models were not simply copying boilerplate text, but for brevity we present ROUGE-F1 as the primary overlap metric in our tables.

As baselines, we use two OpenTSLM variants—SoftPrompt and Flamingo-style cross-attention—and a JoLT-style model with fixed queries. OpenTSLM represents the strongest published baseline, trained end-to-end across multiple time-series datasets, while JoLT is architecturally closer to TS-QFormer in that it routes time-series features through a set of learned query vectors before handing them to a frozen language model. In practice, reproducing these baselines required substantial engineering work. The OpenTSLM repository contained several silent shape mismatches and undocumented preprocessing assumptions (e.g., padding conventions and channel ordering), which we had to resolve to obtain stable forward passes on HAR-QA-CoT. For HAR-QA-CoT, we used the released OpenTSLM checkpoints and adapted their evaluation script to log accuracy and ROUGE metrics under our prompting scheme. By contrast, JoLT provided no README for time-series experiments and several unused or broken modules; we were unable to obtain pretrained checkpoints and therefore trained a JoLT-style model from scratch on HAR-QA-CoT using our own data pipeline.

Hardware and Timing. All experiments were conducted on a single AWS `g5.xlarge` instance equipped with an NVIDIA A10G GPU (24 GB VRAM) and 4 vCPUs. Over the course of the project we consumed approximately 139 GPU-hours in total (about \$140 at \$1.006/hour), including failed runs, debugging, and ablations. A single TS-QFormer training run on HAR-QA-CoT, from initialization to convergence, typically occupied several GPU-hours, with validation sweeps on the HAR test split taking on the order of minutes per model. In contrast, running OpenTSLM inference on the full ECG-QA-CoT test set with the released checkpoints required close to a full day of wall-clock time on the same hardware. This large gap in inference cost is a key reason why we restricted ECG-QA-CoT to zero-shot evaluation only, rather than attempting full fine-tuning in this domain.

In-Domain Results on HAR-QA-CoT. Table 1 summarizes in-domain performance on HAR-QA-CoT. The OpenTSLM variants achieve the strongest overall results, with SoftPrompt slightly outperforming Flamingo-style cross-attention in both accuracy and ROUGE-F1. Our TS-QFormer model outperforms the JoLT-style baseline, but remains noticeably behind the OpenTSLM baselines.

Quantitatively, TS-QFormer narrows the gap to OpenTSLM in terms of ROUGE-F1 while still lagging substantially in accuracy. We observe that TS-QFormer often produces partially correct, fluent answers that capture the right high-level activity (e.g., “walking upstairs” vs. “walking”) but miss fine-grained distinctions required for exact match. Compared to JoLT, our Q-Former adaptor appears more effective at extracting task-relevant temporal features from the frozen encoder: JoLT tends either to underfit (short, generic answers) or to copy fragments of the prompt, resulting in lower accuracy and slightly lower ROUGE-F1. These patterns suggest that while end-to-end training across multiple datasets (as in OpenTSLM) still yields the best HAR performance, a parameter-efficient Q-Former adaptor can recover a nontrivial fraction of that performance without retraining the full encoder–decoder stack.

Out-of-Domain Results on ECG-QA-CoT. To assess cross-domain generalization, we evaluate all models in a strict zero-shot setting on ECG-QA-CoT: no model is fine-tuned on ECG data; instead, we directly reuse the HAR-trained checkpoints and only replace the input time series and question templates. Results are shown in Table 2.

Table 1: Performance on in-domain HAR-QA-CoT data. Acc denotes exact-match answer accuracy; ROUGE-F1 computed over answer strings.

Model	Acc	ROUGE-F1
OpenTSLM-SoftPrompt	0.64	0.55
OpenTSLM-Flamingo	0.62	0.51
JoLT (fixed queries)	0.15	0.49
TS-QFormer (ours)	0.18	0.53

Table 2: Performance on out-of-domain ECG-QA-CoT data (zero-shot). Acc and ROUGE-F1 computed over answer strings.

Model	Acc	ROUGE-F1
OpenTSLM-SoftPrompt	0.007	0.21
OpenTSLM-Flamingo	0.005	0.18
JoLT (fixed queries)	0.000	0.10
TS-QFormer (ours)	0.000	0.11

All models fail dramatically in this regime: answer accuracy is effectively zero for every method, and ROUGE-F1 scores are low across the board. This is not entirely surprising given the substantial domain shift: ECG-QA-CoT involves 12-channel clinical waveforms rather than 3-channel wearable signals, longer temporal contexts (12s vs. 2.5s), and a broader and more fine-grained set of question types. Qualitatively, we find that JoLT and TS-QFormer often generate plausible-sounding chains of thought or generic medical-sounding text but either omit a clear final answer or hallucinate an answer unrelated to the ECG. In our evaluation script, such outputs are treated as incorrect, yielding an accuracy of 0 despite non-zero ROUGE-F1 from superficial token overlap.

The OpenTSLM models fare slightly better in ROUGE-F1, reflecting occasional partial matches to the reference answers, but their accuracy remains near zero as well. Overall, these results highlight that none of the current architectures—including TS-QFormer—achieves meaningful out-of-domain generalization from HAR to ECG under strict zero-shot transfer. From a research perspective, this negative result is informative: it suggests that cross-modal time-series-to-language alignment learned on everyday activities does not trivially carry over to clinical morphology reasoning, and that explicit ECG-domain pretraining or multi-domain training is likely necessary. An example of the prompt and expected response format used in our experiments is provided in Appendix A.

6 Code Overview

A description of each of the files that we have changed in the code is given in Appendix B.

7 Timeline

Our team of three worked approximately 2–3 hours per person per week over a 10-week period. The project followed a structured progression: (i) early weeks focused on problem definition, dataset integration, and baseline reproduction; (ii) mid-phase weeks implemented TS-QFormer, including channel projection, question-conditioned querying, and frozen-LLM integration; and (iii) final weeks completed HAR-CoT training, ECG-CoT zero-shot evaluation, ablations, error analysis, and preparation of the poster and final report. A detailed week-by-week breakdown with responsibilities and effort allocation is provided in the Appendix 3.

8 Research Log

Our project began with an ECG-centric plan: reproduce OpenTSLM and JoLT baselines on ECG-QA-CoT and train TS-QFormer for clinical reasoning. Early work focused on setting up the OpenTSLM codebase and ECG pipelines, but we quickly encountered a major bottleneck—running inference on the full ECG test set took nearly a full day on a single A10G GPU. Retraining baselines on small ECG subsets proved impractical due to long wall-clock times, memory pressure, and concerns about sampling bias. These constraints indicated that an ECG-only study would not yield credible or reproducible comparisons.

This led to a deliberate pivot: we adopted HAR-QA-CoT as our primary in-domain benchmark and reserved ECG-QA-CoT solely for zero-shot evaluation. This shift required reworking our data pipelines, prompt templates, and evaluation plan, but it provided a stable regime in which we could systematically study TS-QFormer under matched compute budgets.

Once the dataset pivot was completed, our focus shifted to implementing and stabilizing TS-QFormer. We reproduced minimal OpenTSLM baselines on HAR to verify our pipelines, then integrated MOMENT-large with a Q-Former adaptor and a frozen OPT-2.7B decoder. Much of the development time was spent resolving technical issues: shape mismatches between encoder and adaptor outputs, sensitivity to learning-rate choices, and formatting inconsistencies in question–CoT–answer sequences that caused several early runs to diverge. A substantial fraction of our compute use went toward these short exploratory runs that exposed configuration bugs and guided us toward a stable training recipe.

With a reliable HAR setup in place, we conducted controlled comparisons across baselines using answer accuracy and ROUGE. For ECG-QA-CoT, we chose a strict zero-shot evaluation: given our compute budget, a single noisy fine-tuning run would have offered limited insight compared to a clean transfer setting. Instead, we allocated remaining resources to improving HAR-side robustness and performing qualitative error analysis on ECG to understand failure modes such as hallucinated activity labels and missed morphological cues.

In summary, the project evolved from an ambitious ECG-heavy proposal into a streamlined study centered on HAR with ECG used as a stress test for cross-domain generalization. Several initially planned experiments (full ECG fine-tuning, larger backbones, multi-seed sweeps) were dropped in favor of a coherent and reproducible methodology. The pivot ultimately enabled a deeper understanding of TS-QFormer’s behavior under realistic resource constraints.

9 Conclusion

We presented TS-QFormer, a parameter-efficient adaptor that enables question-conditioned reasoning over high-frequency multivariate time series by querying a frozen time-series encoder and passing compact, evidence-bearing summaries to a frozen LLM. Empirically, TS-QFormer recovers a substantial fraction of rationale quality (ROUGE) achieved by heavier, end-to-end baselines on HAR while requiring far fewer trainable parameters, and it outperforms a fixed-query JoLT-style design. However, strict zero-shot transfer to ECG fails for all methods tested, highlighting a clear need for ECG-specific pretraining or multi-domain alignment to handle clinical morphology. Our analysis shows that careful channel-aware fusion, question conditioning, improves performance under tight compute budgets. Future work will expand multi-domain pretraining, run multi-seed and ablation studies to quantify robustness, and explore larger LLM/time-series encoder pairings to test scalability of the adaptor approach. Overall, TS-QFormer demonstrates a practical, resource-frugal path toward making LLM-based temporal reasoning useful in low-data scientific and clinical settings.

10 Thought-Experiment on Compute

All experiments were run on an AWS g5.xlarge with a single NVIDIA A10G GPU. Across baseline reproduction (OpenTSLM, JoLT), TS-QFormer implementation, debugging, and HAR-CoT training, we used approximately 139 GPU-hours (\$140 at \$1.006/hour). ECG-QA-CoT was evaluated strictly zero-shot and therefore contributed negligible additional cost.

Compute limits shaped our methodology: we fixed a single backbone (OPT-2.7B), one query-token configuration, and one seed; avoided full hyperparameter sweeps; and skipped ECG fine-tuning due to its long wall-clock time. These constraints prevented variance analysis, larger ablations, and scaling studies, but enabled us to obtain one stable and reproducible configuration for each model family.

Hypothetical Scenario with \$450 Additional Credits. An extra \$450 (447 A10G-hours) would enable: (i) multi-seed runs and systematic sweeps over query counts, LR schedules, and LoRA ranks; (ii) full ECG-QA-CoT fine-tuning for in-domain vs. zero-shot comparisons; and (iii) scaling experiments with larger LLMs and higher-resolution time-series encoders. This would improve statistical confidence, broaden empirical coverage, and clarify how TS-QFormer scales and generalizes across modalities.

A Sample prompt and output from the model

Prompt setup:

1. **Task:**Classify activity from 3-axis accelerometer data
2. **Instructions:** Analyze time series, think step-by-step, reveal label only in final sentence
3. **Possible Labels:** biking, lying, running, sitting, standing, walking, walking_down, walking_up

Sample response from model:

The accelerometer data over the 2.56 s window shows significant variations across all three axes, with repeated peaks and troughs indicating dynamic motion. The intensity is moderate and consistent, suggesting walking rather than high-impact running or near-static postures. The pattern is more regular and downward-directed than purely horizontal or vertical movement, consistent with descending steps. **Answer: walking_down.**

B Code Overview

Description: Our system uses a Q-Former to bridge time series encoders with large language models, supporting both representation learning (RL) and generative learning (GL) training paradigms for ECG diagnosis and human activity recognition.

Code Attribution: Code Taken from Existing Sources

- models/blip2.py (Lines 1-85): Base BLIP-2 framework from Salesforce
- models/bert.py: BERT implementation for Q-Former from Hugging Face and BLIP-2
- models/encoders.py: Time series encoder interfaces adapted from MOMENT
- models/jolt_representation_learning.py: Base JoLT RL implementation
- models/jolt_generative_learning.py: Base JoLT GL implementation
- ts_datasets/load_ptbxxl.py and load_mitbih.py: Standard ECG dataset loaders
- train/trainer.py: Base training loop structure

Original Contributions:

1. Q-Former Modifications for Time Series (models/jolt_representation_learning.py)
 - Lines 45-60: Q-Former vocabulary resizing to match tokenizer
 - Lines 62-66: Query token weight initialization from base BERT
 - Lines 148-157: Runtime vocabulary validation and clamping
2. Channel Aggregation with Linear Projection (models/jolt_representation_learning.py)
 - Lines 29, 73-84: Configuration and initialization of learnable channel projection
 - Lines 109-123: Dynamic channel aggregation (mean vs linear projection)
3. Representation Learning (RL) Training Modifications (models/jolt_representation_learning.py)
 - Lines 97-230: Modified forward pass with three-stage contrastive learning (ITC, ITM, LM)
4. Generative Learning (GL) Training Modifications (models/jolt_generative_learning.py)
 - Lines 75-77: Decoder projection layer (Q-Former to LLM hidden size)
 - Lines 115-200: Instruction tuning pipeline with prompt masking
5. Training Infrastructure Enhancements (train/trainer.py)

- Lines 22-50: Added early stopping functionality
- Lines 145-220: Added gradient accumulation, mixed precision, WandB logging

6. Dataset Implementations

- `ts_datasets/har_cot_text_pair_dataset.py` (entire file): HAR with Chain-of-Thought reasoning
- `ts_datasets/ecg_qa_cot_text_pair_dataset.py` (entire file): ECG question-answering dataset

C Timeline Details

Phase	Approx. Dates	Key Activities & Deliverables	Effort (person-hours)
Project Setup & Proposal	Late Oct – Nov 7	Defined TS-QA task; surveyed OpenTSLM and JoLT; scoped TS-QFormer as a parameter-efficient adaptor; integrated HAR-CoT/ECG-CoT formats; drafted project proposal.	12–18
Baseline Reproduction & Data Pipelines	Nov 1 – Nov 20	Implemented data loaders; reproduced OpenTSLM-SoftPrompt, Flamingo-style baseline, and JoLT; established unified evaluation harness (accuracy + ROUGE).	12–18
TS-QFormer Implementation	Nov 20 – Nov 25	Implemented channel projection, question-conditioned Q-Former, and OPT-2.7B interface; validated adaptor behavior; switched training dataset to HAR-CoT after ECG-CoT feasibility analysis.	12–18
TS-QFormer Training & In-Domain Evaluation	Nov 25 – Dec 5	Ran alignment (Stage 1) and supervised generation (Stage 2) on HAR-CoT; conducted HAR-QA-CoT in-domain evaluations; benchmarked against all baselines.	12–18
Out-of-Domain Evaluation, Analysis, Poster & Report	Dec 6 – Dec 11	Ran ECG-CoT zero-shot evaluation; performed qualitative error analysis; prepared poster figures and examples; drafted final report.	12–18
Total		10 weeks of coordinated work by 3 members	60–90

Table 3: Week-by-week project timeline, milestones, and approximate effort.

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